

ASGA™ Architecture Map

Autonomous Systems Governance Architecture

Official Autonomous Operational Governance Reference Map for Autonomous Systems, AI Agents, Robotics, Autonomous Vehicles, Industrial Automation, Critical Infrastructure, Human-AI Environments, and Future Autonomous Ecosystems

ASGA™ serves as the official autonomous operational governance reference map used to identify, classify, navigate, validate, and understand autonomous operational spaces across autonomous systems, AI agents, robotics, industrial automation, cyber-physical systems, Human-AI environments, critical infrastructure, and future autonomous ecosystems.

The purpose of ASGA™ is not to execute autonomous operations.

The purpose of ASGA™ is to identify, define, and map the governable autonomous operational spaces of autonomous systems.

ASGA™ serves as the official autonomous operational governance reference map for:

- Autonomous Systems
- AI Agents
- Robotics
- Autonomous Vehicles
- Multi-Agent Systems
- Human-AI Environments
- Industrial Automation
- Critical Infrastructure
- Cyber-Physical Systems
- Future Autonomous Ecosystems

Core Architectural Principle

Every autonomous operational space answers one primary question.

No autonomous operational space duplicates the purpose of another autonomous operational space.

Every autonomous operational space begins where the previous autonomous operational space ends.

Together these autonomous operational spaces form the official autonomous operational governance reference map of ASGA™.

Official Governance Flow

Identity → Authority → Execution → Constraints → Risk & Trust → Coordination → Escalation → Accountability → Compatibility → Evolution

1. Autonomous Identity Standard (AIS)

Governed Space: Autonomous Operational Identity

Core Question

Who is the autonomous operational actor?

Canonical Definition

AIS defines the conditions under which an autonomous operational actor becomes uniquely identifiable, attributable, verifiable, and governable.

Governance Boundary

Begins when an autonomous operational actor exists.

Ends when autonomous operational identity has been established and verified.

2. Autonomous Authority Standard (AAS)

Governed Space: Autonomous Operational Authority

Core Question

Who has authority to perform autonomous operations?

Canonical Definition

AAS defines the conditions under which autonomous operational authority is established, delegated, validated, restricted, and governed.

Governance Boundary

Begins when autonomous operation requires authority.

Ends when operational authority has been legitimately established.

3. Autonomous Execution Standard (AES)

Governed Space: Autonomous Execution

Core Question

What autonomous actions may be executed?

Canonical Definition

AES defines the conditions under which autonomous operations are authorized, executed, monitored, controlled, and safely terminated.

Governance Boundary

Begins when autonomous execution is initiated.

Ends when autonomous execution has been completed or terminated.

4. Autonomous Constraint Standard (ACS)

Governed Space: Autonomous Operational Constraints

Core Question

Which operational constraints may never be violated?

Canonical Definition

ACS defines the conditions under which autonomous operational constraints preserve safe, stable, and governable autonomous execution.

Governance Boundary

Begins when operational limitations apply.

Ends when constraint compliance has been evaluated.

5. Autonomous Risk & Trust Standard (ARTS)

Governed Space: Autonomous Operational Risk & Trust

Core Question

What level of operational risk and trust is acceptable?

Canonical Definition

ARTS defines the conditions under which operational risks are identified, evaluated, mitigated, and balanced against trustworthy autonomous operation.

Governance Boundary

Begins when autonomous uncertainty creates operational consequences.

Ends when operational trust has been evaluated against identified risks.

6. Autonomous Coordination Standard (ACOS)

Governed Space: Autonomous Operational Coordination

Core Question

How do autonomous systems coordinate operations?

Canonical Definition

ACOS defines the conditions under which autonomous operational actors coordinate, synchronize, collaborate, and preserve governed operational cooperation.

Governance Boundary

Begins when multiple autonomous operational actors interact.

Ends when coordinated autonomous operation has been established.

7. Autonomous Escalation Standard (AESS)

Governed Space: Autonomous Operational Escalation

Core Question

When and how should autonomous operation escalate?

Canonical Definition

AESS defines the conditions under which autonomous operational escalation becomes necessary, legitimate, controlled, and governable.

Governance Boundary

Begins when autonomous operation exceeds predefined governance conditions.

Ends when escalation has been appropriately governed.

8. Autonomous Operational Accountability Standard (AOAS)

Governed Space: Autonomous Operational Accountability

Core Question

How is accountability preserved throughout autonomous operation?

Canonical Definition

AOAS defines the conditions under which accountability remains attributable, continuous, evidence-based, traceable, and governable throughout autonomous operation.

Governance Boundary

Begins when autonomous responsibility is established.

Ends when accountability can be continuously demonstrated.

9. Autonomous Compatibility Standard (ACPS)

Governed Space: Autonomous Operational Compatibility

Core Question

How do autonomous systems remain operationally compatible?

Canonical Definition

ACPS defines the conditions under which autonomous operational systems remain interoperable, compatible, coordinated, and governable across heterogeneous operational environments.

Governance Boundary

Begins when autonomous systems interact across operational environments.

Ends when operational compatibility has been validated.

10. Autonomous Evolution Standard (AEVS)

Governed Space: Autonomous Evolution

Core Question

How may autonomous systems evolve while remaining governable?

Canonical Definition

AEVS defines the conditions under which autonomous operational systems evolve while preserving governance integrity, accountability, compatibility, operational safety, and trustworthy autonomous operation.

Governance Boundary

Begins when autonomous evolution is proposed or initiated.

Ends when autonomous evolution has been authorized, validated, and fully governed.

Core Governed Spaces

- Autonomous Operational Identity
 - Autonomous Operational Authority
 - Autonomous Execution
 - Autonomous Operational Constraints
 - Autonomous Operational Risk & Trust
 - Autonomous Operational Coordination
 - Autonomous Operational Escalation
 - Autonomous Operational Accountability
 - Autonomous Operational Compatibility
 - Autonomous Evolution
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Architectural Position

ASGA™ governs the complete autonomous operational lifecycle.

The architecture progresses:

Autonomous Identity → Autonomous Authority → Autonomous Execution → Autonomous Constraints → Autonomous Risk & Trust → Autonomous Coordination → Autonomous Escalation → Autonomous Operational Accountability → Autonomous Compatibility → Autonomous Evolution

Together these autonomous operational spaces govern how autonomous systems establish identity, receive authority, execute operations, remain within operational constraints, manage operational risk, coordinate with other systems, escalate exceptional situations, preserve accountability, maintain compatibility, and evolve while remaining continuously governable.

What ASGA™ Defines

ASGA™ defines:

- where autonomous operational identity begins,
 - where autonomous authority is established,
 - where autonomous execution occurs,
 - where operational constraints apply,
 - where operational risks emerge,
 - where autonomous coordination is established,
 - where operational escalation becomes necessary,
 - where accountability is preserved,
 - where operational compatibility is validated,
 - where autonomous evolution remains governable.
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ASGA™ provides the official autonomous operational governance language used for autonomous operational space identification across autonomous systems.

Compatible OOF® Architectures

- GOA™ — Governance Architecture
 - ORA™ — Operational Reality Architecture
 - AGA™ — Accountability Governance Architecture
 - AIG® — AI Governance Architecture
 - CLIA® — Cognitive Governance Intelligence Architecture
 - MGIA™ — Memory Governance Intelligence Architecture
 - RIS™ — Runtime Integrity Standard
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Canonical Principle

ASGA™ does not govern what autonomous systems intend to achieve. ASGA™ governs how autonomous operation itself remains identifiable, authorized, constrained, trustworthy, coordinated, accountable, compatible, evolvable, and continuously governable before, during, and after autonomous execution.

Canonical Closing Statement

ASGA™ serves as the official autonomous operational governance reference map for autonomous systems, AI agents, robotics, autonomous vehicles, industrial automation, critical infrastructure, Human-AI environments, and future autonomous ecosystems. By defining the primary governable autonomous operational spaces of autonomous systems, ASGA™ provides a structured governance reference architecture used to identify, classify, navigate, validate, govern, and understand autonomous operation throughout the complete autonomous operational lifecycle.

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Canonical Source

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